National University of Singapore
School of Computing
CS3244 Machine Learning

Tutorial 4: Neural Networks

Issue: March 1, 2018 Due: March 8, 2018

Important Instructions:

- *Your solutions for this tutorial must be TYPE-WRITTEN.*
- *Make TWO copies of your solutions: one for you and one to be SUBMITTED TO THE TUTOR IN CLASS. Your submission in your respective tutorial class will be used to indicate your CLASS ATTENDANCE. Late submission will NOT be entertained.*
- *YOUR SOLUTIONS TO QUESTIONS* **BL 7** *and* **BL 8c** *will be GRADED for this tutorial.*
- *You may discuss the content of the questions with your classmates. But everyone should work out and write up ALL the solutions by yourself.*

BL 7 On page 20 of the "Neural Networks" lecture slides, I have only provided the derivation of Δw_{hk} for the BACKPROPAGATION algorithm. Complete the derivation of $\Delta w_{ih} = -\eta \partial L_d / \partial w_{ih} = \eta \delta_h x_i$, as shown below in the solution space. Show the steps of your derivation clearly. **No marks will be awarded for not doing so.**

Solution.

$$
\begin{aligned}
\frac{\partial L_d}{\partial w_{ih}} &= \frac{\partial L_d}{\partial o_h} \frac{\partial o_h}{\partial net_h} \frac{\partial net_h}{\partial w_{ih}} \text{ where } net_h = \sum_{i'} w_{i'h} x_{i'} \\
\frac{\partial L_d}{\partial o_h} &= \sum_{k \in K} \frac{\partial L_d}{\partial o_k} \frac{\partial o_k}{\partial net_k} \frac{\partial net_k}{\partial o_h} \\
&= \sum_{k \in K} -(t_k - o_k) o_k (1 - o_k) w_{hk} = -\sum_{k \in K} w_{hk} \delta_k \\
\frac{\partial o_h}{\partial net_h} &= \frac{\partial \sigma(net_h)}{\partial net_h} = o_h (1 - o_h) \\
\frac{\partial net_h}{\partial w_{ih}} &= x_i \\
\Delta w_{ih} &= -\eta \frac{\partial L_d}{\partial w_{ih}} = \eta \left(\sum_{k \in K} w_{hk} \delta_k \right) o_h (1 - o_h) x_i = \eta \delta_h x_i \ .
\end{aligned}
$$

BL 8 In every multi-layer network of perceptron units described below,

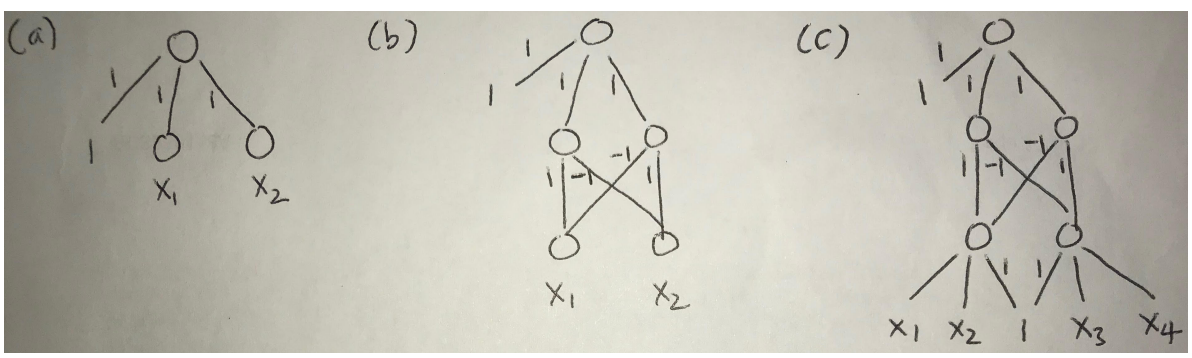
- There should be only one (Boolean) output node and an input node for every (Boolean) input.
- A Boolean is -1 if false, and 1 if true.
- The activation function of every (non-input) node is a -1 to 1 step function, including that of the output node (see page 6 of the "Neural Networks" lecture slides).
- Your weights must be **integers** and kept **small**, but possibly negative.
- Keep your networks as symmetric as possible – doing this in questions a and b may help you in question c.
- You don't have to draw edges with weight 0.

(a) Construct and draw a perceptron network with no hidden layers that implements (x_1 OR x_2).

(b) Construct and draw a perceptron network with one hidden layer (with two nodes) that implements (x_1 XOR x_2).

(c) Combining your constructions in questions a and b, construct and draw a perceptron network with two hidden layers (two nodes each) that implements (x_1 OR x_2) XOR (x_3 OR x_4).

Solution.



TM 4.1 What are the values of weights w_0, w_1, and w_2 for the perceptron whose decision surface is illustrated in figure a on page 7 of the "Neural Networks" lecture slides? Assume the surface crosses the x_1 axis at -1, and the x_2 axis at 2.

Solution. $w_0 = w_1 = -2w_2$, $w_0, w_1 < 0$ and $w_2 > 0$. For example, $w_0 = -2$, $w_1 = -2$, and $w_2 = 1$.

TM 4.3 Consider two perceptrons defined by the threshold expression $w_0 + w_1x_1 + w_2x_2 > 0$. Perceptron A has weight values $w_0 = 1$, $w_1 = 2$, $w_2 = 1$ and perceptron B has the weight values $w_0 = 0$, $w_1 = 2$, $w_2 = 1$. True or false? Perceptron A is *more general than* perceptron B; the notion of *more general than* is previously defined on page 12 of the "Concept Learning" lecture slides.

Solution. True.